

# **BAHRIA UNIVERSITY KARACHI CAMPUS**

## **DEPARTMENT OF COMPUTER SCIENCE**



**Bahria University**  
Discovering Knowledge

## **INTRO TO DATA SCIENCE (CSL-487)**

**Class: 6A – Term: Spring 2024**

### **PROJECT NAME**

**Weather Forecasting**

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## **1. Acknowledgement:**

We would like to express our gratitude to our instructor and mentors for their invaluable guidance and support throughout this project. Their insights and encouragement were crucial in the successful completion of this Weather prediction Application.

## **2. Abstract:**

This report details the development of a Weather Forecasting App using Python. The application retrieves current weather data and forecasts future temperatures for a given city using the OpenWeatherMap API. It utilizes time series analysis and ARIMA modeling for predictions. The application is built with a graphical user interface (GUI) using Tkinter, allowing users to input city names and view weather forecasts graphically.

## **3. Introduction:**

Weather forecasting is a crucial service that aids in daily planning and emergency preparedness. Accurate weather predictions help individuals and organizations make informed decisions. This project aims to develop a user-friendly application that provides current weather data and forecasts future temperatures for any given city.

## **4. Background/Literature Review:**

Weather forecasting has evolved significantly with advancements in technology. Traditional methods relied on historical data and manual calculations. Today, modern techniques use sophisticated algorithms and models like ARIMA (AutoRegressive Integrated Moving Average) to predict future weather conditions based on time series data. This project leverages these advanced methods to enhance the accuracy of weather forecasts.

## **5. Problem statement:**

Existing weather forecasting applications often lack user-friendly interfaces and fail to provide detailed forecasts for multiple future steps. There is a need for an application that not only delivers accurate current weather information but also provides a reliable and visually intuitive forecast for the coming days.

## **6. Objectives and goals:**

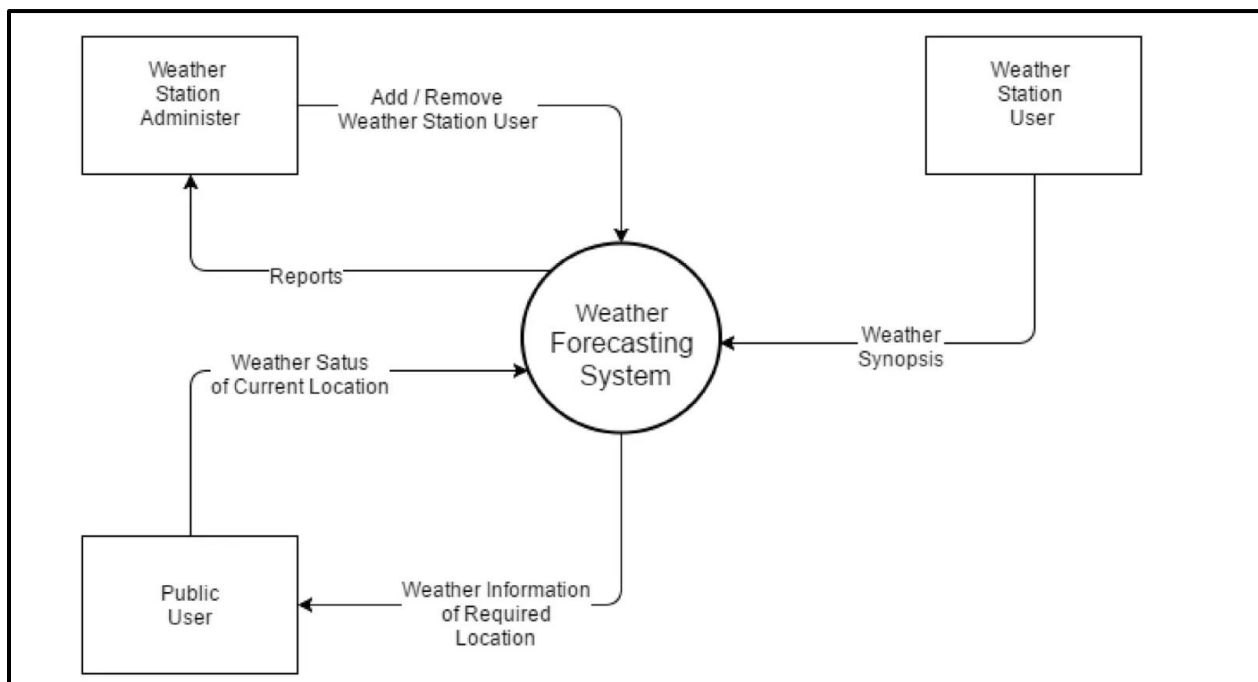
- To develop a GUI-based weather forecasting application.
- To retrieve current weather data for any given city.

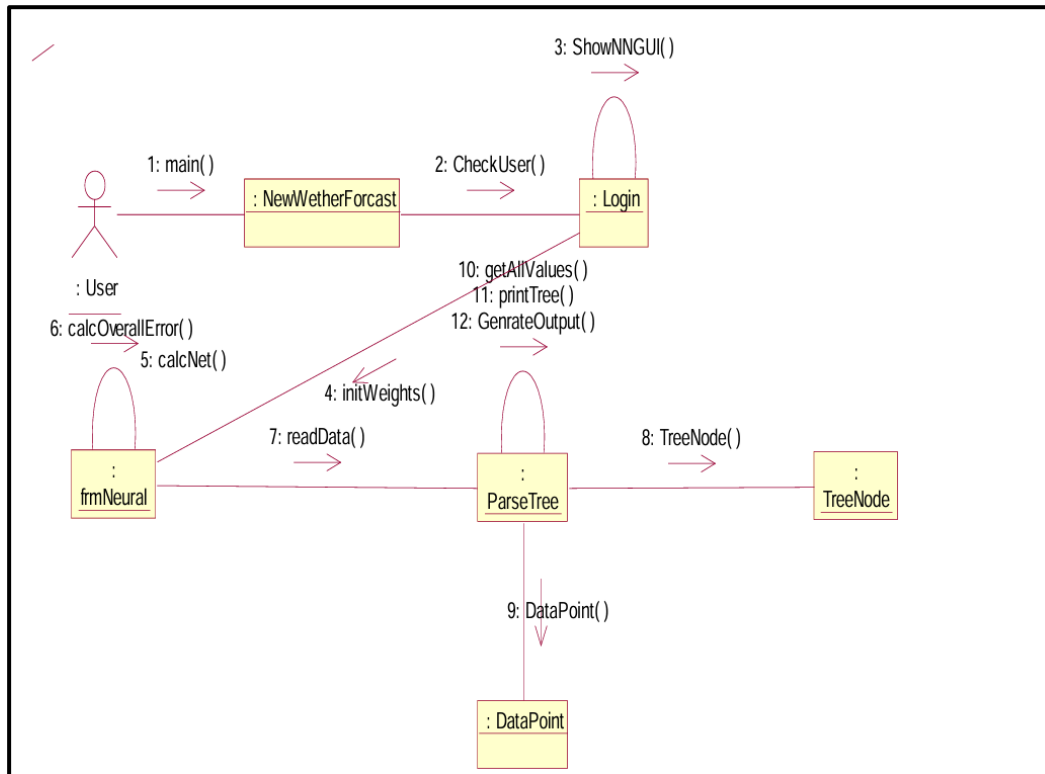
- To forecast future temperatures using ARIMA modeling.
- To provide a graphical representation of the forecasted data.
- To ensure the application is user-friendly and efficient.

## 7. Project Scope:

The project encompasses the development of a desktop application using Python and Tkinter for the GUI. It integrates the OpenWeatherMap API for data retrieval and utilizes time series analysis and ARIMA models for forecasting. The application will be capable of displaying current weather conditions and forecasting future temperatures for a specified city.

## 8. Workflow:





## 9. Overview of project :

The Weather Forecasting App is designed to provide both current weather data and future forecasts. Users can input a city name to get the current temperature and weather conditions. The application also forecasts temperatures for the next few days, displaying the results in a clear and intuitive graphical format.

## 10. Tools and Technologies :

- **Programming Language:** Python
- **Libraries:** requests, pandas, matplotlib, statsmodels, numpy, tkinter
- **API:** OpenWeatherMap
- **GUI Framework:** Tkinter
- **Modeling:** SARIMAX (from statsmodels)

## 11. Project features (functional and non-functional requirements):

### Functional Requirements:

- User input for city name.
- Retrieval of current weather data.
- Forecasting of future temperatures.
- Display of forecasted data in a graphical format.

### Non-Functional Requirements

- User-friendly interface.
- Efficient data retrieval and processing.
- Accurate forecasting using ARIMA models.

## 12. Data Science implemented concepts:

```
In [1]: import requests
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.stattools import adfuller
import numpy as np
import tkinter as tk
from tkinter import messagebox, scrolledtext
from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg
```

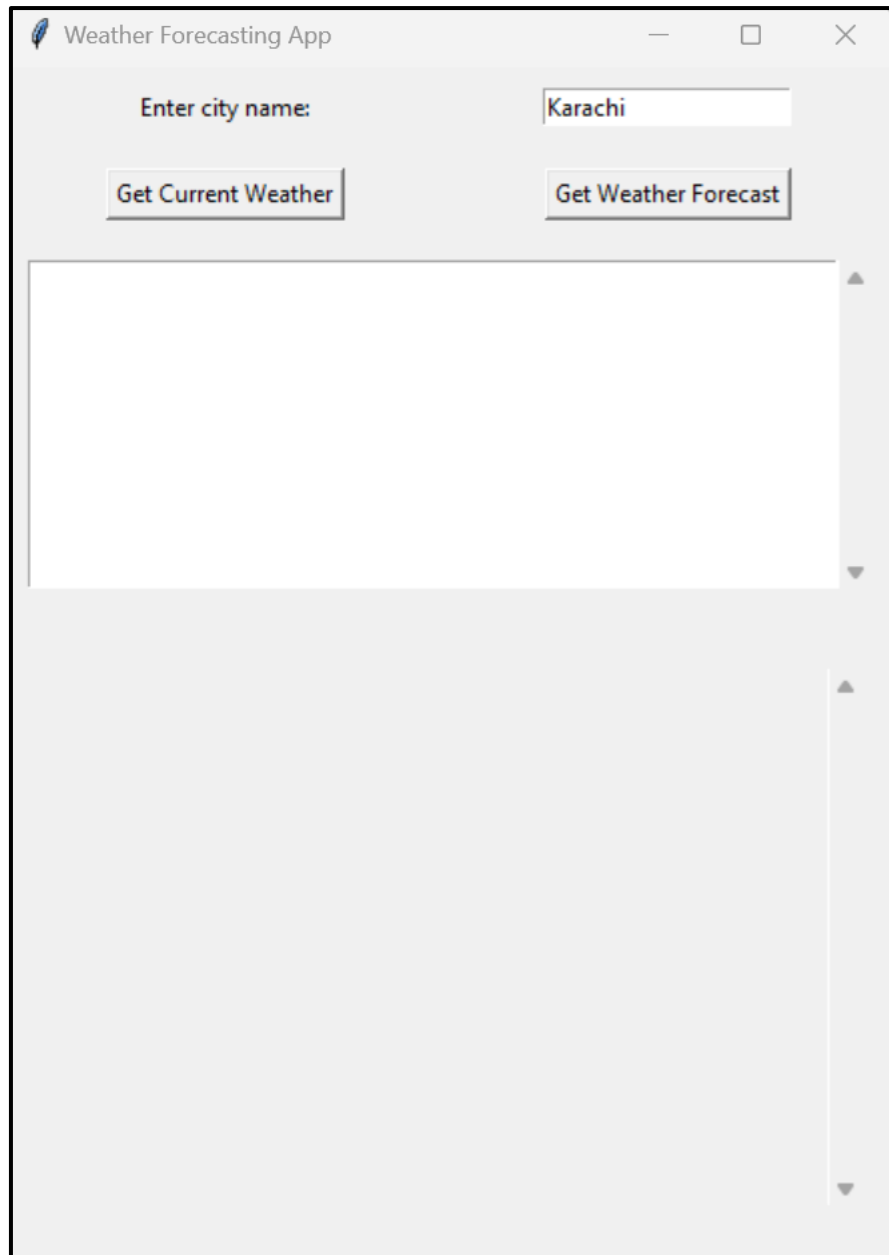
```
In [ ]: def adf_test(series):
    result = adfuller(series, autolag='AIC')
    speak('ADF Statistic: {}'.format(result[0]))
    speak('p-value: {}'.format(result[1]))
    for key, value in result[4].items():
        speak('Critical Values:')
        speak(' {}: {}'.format(key, value))

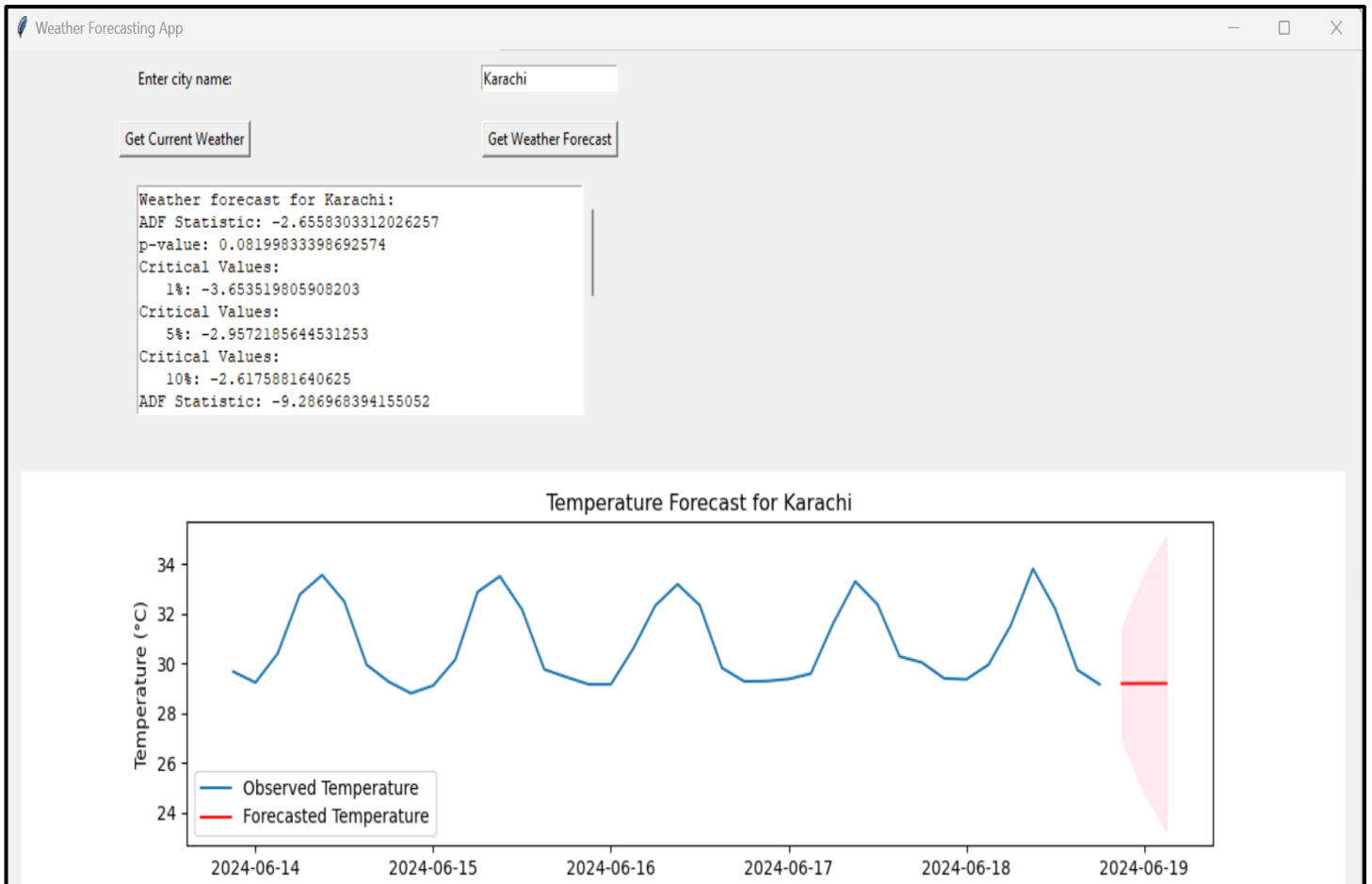
def difference_series(series):
    return series.diff().dropna()

def fit_arima_model(series, order=(1,1,1)):
    model = SARIMAX(series, order=order)
    model_fit = model.fit(dispatch=False)
    return model_fit

def predict_future_weather(model, steps=3):
    forecast = model.get_forecast(steps=steps)
    return forecast.predicted_mean, forecast.conf_int()
```

### 13. Output:





## 14. Conclusion:

The Weather Forecasting App successfully retrieves current weather data and forecasts future temperatures using ARIMA modeling. The user-friendly interface and graphical representation of data make it an effective tool for users to plan their activities based on weather predictions. Future enhancements could include extended forecast capabilities and additional weather parameters.



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